Vraj Patel

Engineering student at Queen's University

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EDUCATION

Queen's University Sep 2026 - Apr 2030 (expected)
BASc, Engineering | Kingston, ON

EXPERIENCE

Founder & Captain, FTC Teams #27964 & #32514 Sep 2023 - Jun 2026
William Lyon Mackenzie CI, Toronto, ON

- Founded a school robotics program with a general robotics club, appointed technical leads, and led 50+ members across two FTC teams.
- Led the FTC mechanical sub-team, designing the mecanum drivetrain and manipulator arm in Fusion 360 and assembling the school's first competition robot in two months.
- Led weekly Arduino workshops and school outreach, introducing 100+ students to electronics and programming.

Database Designer (Co-op) Sep 2024 - Feb 2025
Access Group Inc., Toronto, ON

- Automated Intune, Defender, and remote-monitoring reports with PowerShell and Power BI, replacing manual Excel reports.
- Built Power BI dashboards from ConnectWise API data, giving IT staff access to account information for 100 users.

PROJECTS

ESP32 Quadcopter | C++, ESP32 Jul - Aug 2026

- Built a transmitter around an F310 gamepad and ESP32-S3, sending throttle, roll, pitch, yaw, and arm/disarm commands over ESP-NOW.
- Assembled an F450 quadcopter with an ESP32 flight controller running esp-fc for takeoff, hover, and manual flight.
- Wired an MPU-6500 IMU, BMP280 barometer, and QMC5883L magnetometer to the quadcopter's ESP32 over I2C for motion, pressure, and heading measurements.

6-DOF Robotic Arm | Python, Arduino Jan - Mar 2026

- Designed and assembled a 3D-printed arm with six servos, two at the shoulder, and a NEMA-17 stepper base.
- Programmed the arm to reach typed 3D coordinates using Python inverse kinematics, keeping each joint within its limits.
- Developed an interface with sliders and a 3D arm view to control the arm over a serial link to Arduino.

Guitar Effects Pedal PCBs | KiCad Sep - Nov 2025

- Created KiCad layouts and Gerber files for Boss DS-1, Big Muff Pi, and EHX LPB-1 pedals from existing schematics.
- Soldered and debugged three working through-hole pedals, applying gain staging, clipping, and biasing.

AWARD

1st Place in Canada - CyberTitan 2026
National cybersecurity competition; six-person team; Windows system hardening.

SKILLS

Design: KiCad, Fusion 360, SolidWorks, Onshape
Hardware & fabrication: ESP32, RP2040, STM32, Raspberry Pi, soldering, 3D printing
Programming: C, C++, Python, PowerShell
Controls: PID control, robot kinematics
Software tools: Linux, Git/GitHub